

# ATISHAY JAIN

## FULL-STACK SOFTWARE ENGINEER

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### EDUCATION

**B.Tech, Computer Engineering - Thapar Institute of Engineering and Technology** | Aug 2023 - Jun 2027

CGPA: 8.0/10.0 | Patiala, Punjab, India

### EXPERIENCE & LEADERSHIP

**Def-Space Summer Internship Programme - Bharat Space Education Research Centre (BSERC)** | Jun 2026 - Aug 2026

- Completed a 120-hour, 6+ week Def-Space programme focused on defence and space technologies and responsible innovation.
- Earned BSERC-DSI-2026-2601 after completing structured learning on defence-space systems and responsible innovation.

**Finance & Documentation Manager - TEDxTIET** | Aug 2025 - Feb 2026

- Managed a INR 3.5 lakh event budget through allocation, expense tracking, sponsor records, and variance reporting.

### PROJECTS

**ReceiptSplit - Multi-room Bill-Splitting Platform | 2026**

*Next.js, TypeScript, FastAPI, async SQLAlchemy, PostgreSQL, Docker, SSE, OCR*

- Built a multi-room bill-splitting MVP with OCR drafts, deterministic split rounding, durable room events, and stateful settlement workflows.
- Implemented authenticated room/member flows, auditable settlement states, and UPI link/QR coordination with manual payment confirmation by design.
- Expanded frontend regression checks from 51 to 82 (+61%) and maintained 379 backend test functions across 53 files for authenticated room/member and settlement flows.

**TIET-LOOP - Campus Event Platform | 2025 - 2026**

*React, Vite, Tailwind CSS, FastAPI, WebSockets, Google Maps*

- Collaborative Full-Stack Contributor to a campus platform with 8 persistence models spanning authentication, profiles, discovery, RSVPs, friends, chat, carpools, maps, recommendations, and admin workflows.
- Contributed React/FastAPI integration and real-time WebSocket flows; signed JWT access tokens expire after 30 minutes and passwords use Argon2 hashing.
- Implemented map/list event discovery, PWA support, RSVP and carpool workflows, and Google Maps deep links for campus event coordination.

**Hybrid GenAI Transaction Categorizer | 2025 - 2026**

*Python, FastAPI, React, ONNX Runtime, DistilBERT, SQLite, local LLM integration*

- Built a 3-path offline categorization pipeline - merchant rules, ONNX DistilBERT, then local-LLM fallback - over 1,600 labeled transaction rows across 10 categories.
- Reduced local CPU fallback latency from 60-80 seconds to 5-13 seconds (78-94% lower) by consolidating Qwen2.5, shortening prompts, and capping output at 40 tokens; passed 27 local tests.
- Added a feedback-controlled merchant-memory endpoint and 0.90 confidence gate, avoiding uncontrolled model updates while keeping decisions auditable.

**Mahoraga - Adaptive RL Environment | 2026**

*Python, Gymnasium, FastAPI, Gradio, Qwen 2.5, LoRA*

- Contributed as a Backend / RL Systems Engineer on a 5-action adaptive environment with inspectable reward components, Gymnasium support, and LLM/backend integration.
- Meta OpenEnv Hackathon 2026 Finalist; sped up T4 fine-tuning by ~3x (1,024 to 512 token context; batch size 2 to 4) and verified 145 Python checks.
- Exposed reset, step, and model-status APIs for a React dashboard with manual controls and deliberate rule-based fallback when LoRA weights are absent.

### SKILLS

**Languages:** Python, TypeScript, JavaScript, C++, C, SQL

**Full-Stack:** React, Next.js, Vite, Tailwind CSS, FastAPI, REST APIs, WebSockets, JWT, Google Maps

**Data & Infrastructure:** PostgreSQL, Supabase, SQLite, SQLAlchemy, Docker, Git, GitHub Actions, Render

**Applied AI:** ONNX Runtime, DistilBERT, Transformers, local LLM integration, Gymnasium, LoRA

### CERTIFICATIONS

BSERC Def-Space Summer Internship Programme (120 hours) | NVIDIA Foundations of Deep Learning | Duke University - Introduction to RAG | CI/CD Pipeline with Docker